PeaPod for XYZ - Requirements

Outlining the Implementation-Specific Requirements for a PeaPod XYZ

Extends: ***PeaPod - Requirements***

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1 Introduction

1.1 Purpose

The purpose of this document is to outline both the category requirements (Section 2.2) for an implementation of the PeaPod framework (See *PeaPod - Requirements*) that XYZ and the scoped requirements (Section 1.3) for the design being proposed by PeaPod Technologies Inc.: **PeaPod for XYZ**.

1.2 Design Paradigm

All Solutions provided by PeaPod Technologies are fully documented and backed by our rigorous top-down engineering-design paradigm, applied at all steps, enabling us to frame the Problem in such a way that a Solution is a provable and self-evident combination of our Services:

- **2.1 - Problem Statement**: A scoped overview of the opportunity or **Problem**. Anything not covered by this **Statement** is not considered in the execution of the project.
- **2.2 - Solution Requirements**: Categorical requirements for *any* solution to the problem, interpolated from the scope boundaries in the **Problem Statement**. If any of these are not met, the problem is not solved, providing a distinct *pass/fail threshold*, or "razor", for possible Solutions.
- **2.3 - Stakeholders and Values**: *Perspectives* in consideration (i.e. ITCGI, the client, client-of-client), along with a summarized *value proposition* for each person or group, derived from the **Problem Statement** and **Requirements**. These are the people who are and will be *directly affected* by the **Problem**, and as such their **Values** will influence the selection of factors about which we shall select the ideal **Solution**.
- **2.4 - Problem-Solving Goals**: *Conceptual* design goals (e.g. Safety, Efficiency) derived from **Requirements** and **Stakeholder Values**.
- **2.5 - Solution Objectives**: *Tactical* implementation targets derived from the **Requirements** and **Goals**.

- **2.6 - Metrics**: Granular, *quantitative* measures of design success/fit/utility/etc. derived from the **Objectives**, which are either Constrained or Graded according to the **Requirements**, **Goals**, and **Objectives**.
- **2.7 - Constraints**: *Mandatory* thresholds (i.e. true/false, pass/fail) and extrema (minima, maxima) for evaluating and disqualifying proposed solutions along Constrained **Metrics**.
- **2.8 - Criteria**: *Points-based* system for evaluating and ranking proposed solutions along Graded **Metrics**.

1.3 Scope and Justification

SC1. Lorem ipsum dolor sit amet, consectetur adipiscing elit:

  SC3a. Sed auctor, nunc nec ultricies ultricies, nunc nunc ultricies nunc, nec ultricies nunc nunc nec.

1.4 Definitions

A number of useful definitions have emerged from the above scoping:

1. **ABC** - Lorem ipsum dolor sit amet, consectetur adipiscing elit.

2 Framing

2.1 Problem Statement

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed auctor, nunc nec ultricies ultricies, nunc nunc ultricies nunc, nec ultricies nunc nunc nec.

2.2 Solution Requirements

The following are the overall challenge requirements compiled from A, B, and an excerpt from C:

R1. **Must** lorem ipsum dolor sit amet, consectetur adipiscing elit:

   (a) **Should** lorem ipsum dolor sit amet, consectetur adipiscing elit;

2.3 Stakeholders and Values

S1. A - Values, etc.

S2. B - DfX, etc.

2.4 Problem-Solving Goals

HL1. Goal ABC (S1, R1, R1a)

2.5 Solution Objectives

LL1. Objective ABC (HL1)

2.6 Metrics

#	Metric		Units
M1	Metric ABC	(LL1)	Yes/No
M2	Metric XYZ	(LL1)	0 - 100%

2.7 Constraints

Metric	Constraint	Justification
M1	Yes	(S1)

2.8 Criteria

Metric	Criteria	Justification
M2	Should Maximize	(R1a)

3 Reference Designs

3.1 Reference Design XYZ

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